

Notes: Garfinkel & Hammoudeh (RFS, 2025)

Competition and Innovation Revisited: A Project-Level View

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	A motivating manufacturing exercise	3
2.2	Pharma project-level data	4
2.3	Therapeutic markets and competition	4
2.4	BTD shocks	4
2.5	Technology overlap and distance to frontier	4
2.6	Phase-II analysis sample	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Manufacturing benchmark: competition and innovation with aggregation	5
3.2	Event-study regression for BTD announcement returns	5
3.3	Firm-level innovation responses to BTD shocks	6
3.4	Main hazard specification	6
3.5	Testing the inverted-U: interaction with competition	6
3.6	Distance-to-frontier hazard: SM versus SMST	7
3.7	Parallel-trends event study	7
3.8	Reallocation margins	7
4	Identification and instruments (what makes the estimates credible)	8
4.1	Why the standard literature is hard to interpret	8
4.2	Why BTD entry is a plausibly exogenous unlevering shock	8
4.3	Why distance to frontier is well measured here	8
4.4	Parallel trends, anticipation, and sample checks	9
5	Tables and figures	9
5.1	Table 1: Literature survey on competition and innovation	9
5.2	Table 2: Competition-innovation relationship and the influence of firm scope	10

5.3	Figure 1: Drug-development process and the timing of BTB applications	10
5.4	Table 3: Announcement returns of BTB firms, rivals, and controls	10
5.5	Table 4: Firm-level innovation after BTB shocks	11
5.6	Table 5: Summary statistics for the phase-II sample	11
5.7	Figure 2: Event-time hazard coefficients for all rival projects	11
5.8	Figures 3 and 4: Event-time hazards for SMST and SM projects	11
5.9	Table 6: Main hazard results	12
5.10	Table 7: Ex ante leveled markets (no approved-for-sale products)	13
5.11	Table 8: Resource reallocation	13
5.12	Table 9: Financial outcomes after BTB shocks	14
6	Short summary	14
7	Conclusion	15
7.1	Core conclusion (what the paper establishes)	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** What is the causal relation between competition and innovation once two classic measurement problems are addressed directly: aggregation of innovative activity at the firm level or higher, and bad measurement of distance to the technological frontier?
- **Main empirical idea.** The paper uses BTDA awards in the pharmaceutical industry as exogenous “unleveling” shocks to therapeutic markets. A BTDA identifies a drug that appears substantially better than existing treatments, so it shifts the competitive ordering inside one product market.
- **Why pharma is attractive.** Drug development is observed at the project level; each project targets a well-defined therapeutic area; the timing of development stages is observable; and the shock created by a BTDA is tightly linked to one market rather than to the whole firm.
- **Two headline findings.** First, standard firm-level measures mask the relation between competition and innovation because firms reallocate resources across many projects. Second, at the project level the reaction of rivals follows the inverted-U logic of Aghion et al. (2005): in less competitive markets rivals accelerate innovation after a BTDA shock, while in more competitive markets they decelerate.
- **Distance-to-frontier result.** The shape of the response also depends on whether a rival project uses one of the same technologies as the BTDA-awarded drug. Rivals that are technologically closer to the new frontier (the paper’s SMST projects) are more likely to continue development across the board.
- **Auxiliary message.** The paper first reproduces the aggregation problem in a standard manufacturing setting. Once the sample is split by firm scope, the competition-innovation relation is much clearer for narrow-scope firms than for broad-scope firms.
- **Main contribution.** The paper is a direct empirical test of the mechanism behind the Aghion et al. (2005) inverted-U, and it adds the Hashmi (2013) distance-to-frontier extension in a setting where frontier distance can be measured without relying on price-based productivity measures.

2 Data and measurement (key objects)

2.1 A motivating manufacturing exercise

- **Purpose.** Before turning to pharma, the paper shows why aggregation is a real empirical problem in a standard setting.
- **Data.** U.S. manufacturing firms, 1991–2011, with innovation measured by R&D over assets and citation-weighted patenting.
- **Competition.** Chinese import penetration in U.S. industries, instrumented with Chinese imports into comparable high-income countries.
- **Scope measure.** Hoberg-Phillips firm scope proxies how many product markets the firm operates in. Low-scope firms are less likely to have their innovation response masked by within-firm reallocation.
- **Takeaway.** The inverted-U between competition and innovation is visible among narrow-scope firms, but largely disappears among broad-scope firms.

2.2 Pharma project-level data

- **Core source.** Cortellis Competitive Intelligence provides drug development histories, ownership histories, milestone dates, technologies (target actions), and therapeutic indications.
- **Coverage.** U.S.-market drug projects, 2010q1–2021q4.
- **Broad panel.** 67,912 drug projects developed by 7,486 firms, for 1,884,400 project-quarter observations.
- **Ownership.** Cortellis is merged with SDC Platinum M&A data to time ownership changes and map projects to firms over time.
- **Main unit of analysis later on.** Phase-II drug project-quarter.

2.3 Therapeutic markets and competition

- **Market definition.** Drug indications are harmonized into second-subchapter ICD-10 codes. These ICD-10 therapeutic areas are the paper’s product markets.
- **Number of markets.** 1,580 unique therapeutic markets in the full drug data.
- **Main competition measure.** The log number of drug projects in the therapeutic market in a given quarter.
- **Why not rely on sales concentration only?** Sales-based measures miss pipeline competition, are noisy when drugs span multiple indications, and are contaminated by pricing conventions. The paper nevertheless shows robustness to alternative firm-count and sales-HHI measures.

2.4 BTD shocks

- **Institution.** The FDA introduced the Breakthrough Therapy Designation program in July 2012.
- **Economic interpretation.** A BTB signals that a drug shows substantial improvement over available treatments in that therapeutic area, so the award acts like an exogenous unleveling shock to the market.
- **Sample size.** 426 unique BTB designations, mapped to 487 drug projects in 206 ICD-10 markets.
- **Recipients.** 189 BTB-receiving firms, of which 141 are public and 48 private.
- **Rivals.** Any firm actively developing at least one project in the shocked therapeutic market, but without a BTB-awarded drug in that market.

2.5 Technology overlap and distance to frontier

- **Target actions.** Drugs often work through one or more biological targets or actions. Cortellis reports these technologies.
- **SMST rivals.** Rival projects in the same market that share at least one target action with the BTB-awarded drug. These projects are interpreted as closer to the new frontier.

- **SM rivals.** Rival projects in the same market that do not share any target action with the BTD drug. These are farther from the new frontier.
- **Role in the paper.** This is the operational measure of distance to frontier, and it avoids the standard price/productivity contamination problem.

2.6 Phase-II analysis sample

- **Why phase II?** Phase II is the first serious test of efficacy in humans, it is costly, uncertainty is still high, and trial information is publicly reportable. This is the stage where the rival's continuation decision is economically meaningful.
- **Sample size.** 12,038 phase-II projects, 2,319 firms, and 992 ICD-10 markets, for 915,736 project-quarter observations.
- **Outcome.** Whether the project graduates from phase II to phase III.
- **Treatment timing.** A project becomes treated in the quarter its market receives a BTD shock and remains treated thereafter in the main sample.
- **Exposure measure.** Rival exposure to the shocked market equals the number of the firm's projects in that market divided by its total number of projects.

3 Models and empirical specifications (all equations + interpretation)

3.1 Manufacturing benchmark: competition and innovation with aggregation

The paper's motivating manufacturing exercise can be summarized as

$$Innovation_{it} = \beta_1 ImportComp_{jt} + \beta_2 ImportComp_{jt}^2 + \Gamma X_{it} + \alpha_j + \lambda_t + \varepsilon_{it}, \quad (1)$$

where firm i belongs to industry j and $ImportComp_{jt}$ is instrumented Chinese import competition.

- **Dependent variables.** R&D over assets and citation-weighted patenting.
- **Regressor of interest.** Chinese import penetration and its square.
- **Identification.** Imports from China into other high-income countries instrument for U.S. import competition.
- **Interpretation.** The inverted-U is visible for low-scope firms but weak or absent for high-scope firms, which is direct evidence that aggregation matters even in familiar firm-level data.

3.2 Event-study regression for BTD announcement returns

At the firm level, the paper studies whether BTD awards are surprises using a simple event-study regression of cumulative abnormal returns:

$$CAR_{fe} = \alpha + \beta_1 BTDFirm_{fe} + \beta_2 BTDRival_{fe} + \beta_3 SMST_{fe} + \beta_4 SM_{fe} + u_{fe}. \quad (2)$$

- **Window.** $[-1, +1]$ trading days around the announcement.
- **Interpretation.** A positive CAR for the awardee and a negative CAR for rivals supports the view that BTD entry is informative and unexpected.
- **Empirical role.** This is an identification support exercise, not the main competition-innovation test.

3.3 Firm-level innovation responses to BTD shocks

The firm-level benchmark inside pharma is approximately

$$Innovation_{ft} = \alpha_f + \lambda_t + \beta_1 BTDFirm_{ft} + \beta_2 BTDRival_{ft} + \beta_3 SMST_{ft} + \beta_4 SM_{ft} + \varepsilon_{ft}. \quad (3)$$

- **Outcomes.** Patent dummy, citation-weighted patents, and quarterly R&D scaled by assets.
- **Purpose.** Show that even in pharma, firm-level innovation responses are often muted because firms reallocate effort across projects.
- **Main reading.** Rival firms mostly do not show clear firm-wide increases in innovation after a BTD shock, despite strongly negative stock-market reactions.

3.4 Main hazard specification

The paper’s key estimating equation is the Cox proportional-hazards model:

$$h_{ij}(t) = h_{0j} \exp \left[\beta_1 BTDshock_{jt} + X'_{ijt} \beta + \delta_t \right]. \quad (4)$$

- **Outcome.** The hazard that phase-II project i in therapeutic market j advances to phase III.
- **Analysis time.** Quarters since the start of phase II.
- **Baseline hazard.** Stratified by broader first-subchapter ICD-10 market to ensure comparison with close therapeutic neighbors.
- **Treatment.** $BTDshock_{jt} = 1$ once the market receives a BTD and the focal project is a rival project in that market.
- **Interpretation.** A positive coefficient means the shock accelerates the rival’s continuation to phase III.

3.5 Testing the inverted-U: interaction with competition

The main competition test augments the hazard with market competitiveness:

$$h_{ij}(t) = h_{0j} \exp \left[\beta_1 BTDshock_{jt} + \beta_2 (BTDshock_{jt} \times Competition_{jt}) + \beta_3 Competition_{jt} + X'_{ijt} \beta + \delta_t \right]. \quad (5)$$

- **Key sign.** The paper expects $\beta_2 < 0$.

- **Economic meaning.** The same unleveling shock should stimulate catch-up in less competitive markets but discourage innovation in more competitive markets because post-catch-up rents are smaller there.
- **Implementation.** The paper reports both the interaction specification and split-sample hazards for low- versus high-competition therapeutic markets.

3.6 Distance-to-frontier hazard: SM versus SMST

To distinguish close- and far-from-frontier rivals, the hazard becomes

$$h_{ij}(t) = h_{0j} \exp \left[\beta_{SM} SM_{ijt} + \beta_{SMST} SMST_{ijt} + \theta_{SM}(SM_{ijt} \times Competition_{jt}) + \theta_{SMST}(SMST_{ijt} \times Competition_{jt}) + X'_{ijt}\beta + \delta_t \right]. \quad (6)$$

- **Interpretation.** SMST projects are closer to the new frontier created by the BTB drug, so the Hashmi logic predicts more continuation for them.
- **Expected result.** SMST coefficients should be positive; SM coefficients should continue to show the inverted-U pattern through a negative interaction with competition.

3.7 Parallel-trends event study

To validate identification, the paper estimates event-time hazards of the form

$$h_{ij}(t) = h_{0j} \exp \left[\sum_{k \neq 0} \theta_k EverShocked_{ij} \times 1\{EventTime = k\} + X'_{ijt}\beta + \delta_t \right]. \quad (7)$$

- **Purpose.** Check whether eventual treatment and control projects evolve similarly before BTB arrival.
- **Interpretation.** Pre-shock coefficients near zero support the absence of differential pretrends.
- **Variants.** The paper plots these event-time coefficients for the full sample, low/high competition subsamples, and separately for SM and SMST projects.

3.8 Reallocation margins

The paper also studies two additional margins:

$$h_{it}^{unshocked} = h_0 \exp[\beta_1 HiCompShock_{ft} + \beta_2 HiExpShock_{ft} + \beta_3 HiComp_HiExpShock_{ft} + \dots], \quad (8)$$

for rival continuation of unshocked phase-II projects, and

$$Initiation_{ft} = \alpha_f + \lambda_t + \beta_1 HiCompShock_{ft} + \beta_2 HiExpShock_{ft} + \beta_3 HiComp_HiExpShock_{ft} + u_{ft}, \quad (9)$$

for new project starts.

- **Purpose.** Check whether rivals reallocate effort away from the shocked market and toward other projects or new starts.
- **Interpretation.** The results suggest resources are redirected toward less competitive opportunities after rivals are hit hard by a BTM shock.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the standard literature is hard to interpret

- **Aggregation problem.** Firm-level or industry-level innovation mixes together many product markets, so the measured response can wash out when firms reallocate internally.
- **Distance-to-frontier problem.** Standard measures often rely on revenue productivity or markups, which confound technology with prices.
- **Paper's response.** Use project-level pharma data and a shock that directly reveals a newly superior therapy.

4.2 Why BTM entry is a plausibly exogenous unlevering shock

- **Regulatory setting.** A BTM is awarded when the FDA concludes the drug shows substantial improvement over available therapy for a serious condition.
- **Timing.** BTM awards come early in development, often by phase II, precisely when rivals still have meaningful continuation choices ahead of them.
- **Content.** The award contains information about clinical superiority, not only legal protection.
- **Event-study support.** Awardee firms earn strongly positive CARs and rival firms negative CARs around the announcement.
- **Why this is better than patent-based shocks.** A patent protects a narrow claim and still allows extensive between-patent competition; a BTM instead identifies a clear competitive reordering inside one therapeutic market.

4.3 Why distance to frontier is well measured here

- **Leader versus follower.** The BTM drug becomes the new leader in the market.
- **Close versus far followers.** Rival projects that share target actions with the BTM drug are coded as SMST and are interpreted as closer to the frontier.
- **Advantage over standard settings.** The measure is based on technologies and clinical treatment channels rather than output prices.

4.4 Parallel trends, anticipation, and sample checks

- **Parallel trends.** Event-study hazards show no significant divergence in phase-II continuation before treatment.
- **Anticipation via patents or trial completions.** The results survive restricting to markets less likely to experience BTM awards and to subsamples that might have been anticipated.
- **Sample-matching robustness.** A characteristics-based matched sample strengthens rather than weakens the main results.
- **Other robustness.** The results survive excluding cancer therapies, varying whether the BTM recipient is biotech or big pharma, restricting to ex ante leveled markets, and changing the competition measure.

5 Tables and figures

5.1 Table 1: Literature survey on competition and innovation

Read:

- The main message is not just that estimates differ, but that many studies aggregate across countries or industries and rarely measure distance to frontier.
- It motivates the paper’s two design choices: project-level data and a direct proxy for frontier distance.

Table 1
Sampling of literature on competition-innovation relationship and the potential influence of aggregation and distance-to-frontier measurement problems

Early work = 11 papers. End by 1991.^a Two papers have two different shapes in subsamples.

Shape of relationship	# papers with shape	# countries (across papers)	# industries (across papers)	# papers meas. distance to frontier
U – shaped	1	1	14	0
Inverted – U	1	1	56	0
Positive	2	1, 6	21, 6	1
Negative	4	1, 1, 6, 1	1, 1, 6, 191	1
Zero	5	1, 1, 1, 1, 1	12, 259, 1, 191, 73	0

More recent = 15 papers (7 survey based). Through 2012.^b Two papers have two different shapes in subsamples.

Shape of relationship	# papers with shape	# countries (across papers)	# industries (across papers)	# papers meas. distance to frontier
U – shaped	0			
Inverted – U	7	24, 1, 1, 1, 6, 40, 1	5+, 17, 11, 60, ?, 2, 28	3
Positive	5	1, 18, 1, 17, 1	5+, 5+, 21, 15, 160	2
Negative	5	1, 24, 1, 1, 1	1, 5+, 21, 18, ?	2
Zero	0			

Most recent empirical (post-2012).^c N = 8 papers. 1 paper shows different shapes in subsamples.

Shape of relationship	# papers with shape	# countries (across papers)	# industries (across papers)	# papers meas. distance to frontier
U – shaped	0			
Inverted – U	0			
Positive	4	1, 1, 1, 1	10, 223, 80, 14	2
Negative	5	1, 1, 1, 1, 17	116, 11, 5+, 10, 13	2
Zero	0			

Papers selected (nonrandomly) from output of Google Scholar searches on “Competition, Innovation, Inverted-U, Incumbent Innovation.” Selected papers vary in conclusions on shape of competition-innovation relationship. Chose three periods: pre-inverted-U work; period around workhorse Aghion et al. (2005); approximate 10-year window finishing by 2023. Papers listed in footnotes. “?” indicate no information provided on industry-count. In some cases, the number of industries is replaced by the number of product markets, per the paper’s discussion.

^a See Scherer (1965, 1967), Comanor (1967), Mansfield (1981), Scot (1984), Link and Lunn (1984), Culbertson and Mueller (1985), Levin et al. (1985), Angelmar (1985), Lunn (1986), and Geroski (1990, 1991).

^b See Crepon et al. (1998), Blundell et al. (1999), Bassanini and Ernst (2002), Carlin et al. (2004), Aghion et al. (2005), Tang (2006), Tingvall and Poldahl (2006), Aghion et al. (2009), Artes (2009), Lee (2009), Amable et al. (2010), Alder (2010), Santos (2010), Boone et al. (2011), and Pender and Worter (2011).

^c See Hashmi (2013), Autor et al. (2020), Xu and Gong (2017), Cusolito et al. (2023), Cornea and Ormaghi (2014), Benito et al. (2017), Czarnitzki et al. (2014), and Amable et al. (2016).

Table 2
Competition-innovation relationship and the influence of firm scope

A. Scope sorted into medians

	(1) Full sample	(2) CiteW Patent	(3) R&D/AT	(4) CiteW Patent	(5) R&D/AT	(6) CiteW Patent
Dependent var:	R&D/AT	CiteW Patent	R&D/AT	CiteW Patent	R&D/AT	CiteW Patent
<i>U.S. China import</i>	0.006** (2.063)	0.121*** (2.608)	0.009* (1.935)	0.137** (2.410)	0.001 (0.919)	0.238 (1.295)
<i>U.S. China import</i> ²	–0.0003** (–2.185)	–0.001** (–2.425)	–0.001* (–1.914)	–0.001* (–1.690)	–0.000 (–1.205)	–0.001 (–0.866)
Observations	38,489	33,874	21,545	19,770	16,937	14,097
R-squared	.000	.000	.000	.001	.000	.003

B. Scope sorted into terciles (only top and bottom terciles included and the middle tercile is excluded)

	(1) Low scope (low tercile)	(2) CiteW patent	(3) R&D/AT	(4) CiteW patent
Dependent var:	R&D/AT	CiteW patent	R&D/AT	CiteW patent
<i>U.S. China import</i>	0.010* (1.806)	0.221*** (3.696)	0.000 (0.238)	0.133 (0.548)
<i>U.S. China import</i> ²	–0.001* (–1.874)	–0.001** (–2.251)	–0.000 (–0.753)	–0.002 (–1.183)
Observations	14,985	13,629	11,608	9,077
R-squared	.000	.001	.001	.000

C. Scope sorted into quartiles (only top and bottom quartiles reported and the middle quartiles are excluded)

	(1) Low scope (low quartile)	(2) CiteW Patent	(3) R&D/AT	(4) CiteW Patent
Dependent var:	R&D/AT	CiteW Patent	R&D/AT	CiteW Patent
<i>U.S. China import</i>	0.013* (1.649)	0.211*** (3.523)	0.000 (0.170)	0.196 (0.679)
<i>U.S. China import</i> ²	–0.001* (–1.667)	–0.001** (–2.223)	–0.000 (–0.805)	–0.003 (–1.465)
Observations	11,535	10,456	9,539	7,115
R-squared	.000	.001	.000	.001

The tests in this table examine the effect of Chinese import penetration of U.S. industries on firm-level innovation. The sample consists of annual information on 4,230 firms in 120 4-digit SIC manufacturing industries over the period from 1991 to 2011. The observation level of the sample is firm-year. Data on Chinese import penetration is obtained from Autor, Dorn and Hanson (2013). For each industry-year, we obtain the ratio of U.S. imports, as well as the same ratio for a sample of comparison countries’ summed imports, from China to the same industry’s U.S. market volume in 1991. The results reported in this table are from the second stage of a 2SLS regression. We follow a similar identification strategy to that of Autor, Dorn, and Hanson (2013). We instrument U.S. Chinese imports (and its square) with the Chinese imports for a sample of comparison countries’ (and its square). All regressors and instrumental variables are lagged by 1 year.

5.2 Table 2: Competition-innovation relationship and the influence of firm scope

Read:

- In the full manufacturing sample, Chinese import competition enters positively and its square negatively, consistent with an inverted-U.
- Once the sample is split by firm scope, the pattern survives only for low-scope firms. Among high-scope firms the coefficients become small and insignificant.
- This is the paper's first direct evidence that aggregation within multi-product firms can wash out the competition-innovation relation.

5.3 Figure 1: Drug-development process and the timing of BTM applications

Read:

- The figure clarifies why phase II is the right place to look: development costs rise sharply there, approval uncertainty remains high, and BTM awards arrive early enough to affect subsequent continuation.
- It also explains the paper's emphasis on acceleration of phase-II projects rather than on final approval alone.

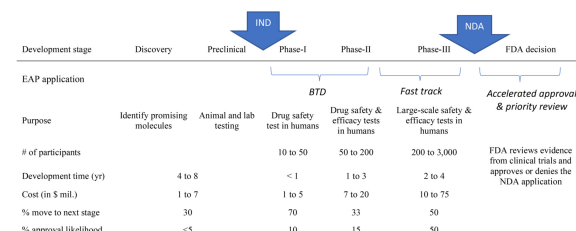


Figure 1
An illustrative description of the drug development process in the United States
For each stage of development, the following information is provided: *EAP* application identifies the recommended development stages for applying to any of the FDA's expedited approval programs. *Purpose* identifies the objective of each development stage. *# of participants* displays the average number of human volunteers in a given clinical stage. *Development time* reports the average number of years a drug project spends in a given stage before moving on to the next one. *Cost* provides the average cost (in \$ millions) associated with a given stage. *% move to next stage* reports the average percentage of drug projects in a given stage that eventually move on to the next one. *% approval likelihood* shows the average probability that a drug project in a given stage eventually receives FDA approval. The two arrows in the first row indicate the timing for IND and NDA applications. IND is an abbreviation for investigational new drug application, and NDA is an abbreviation for new drug application.

	Full sample (1)	Full sample (2)	Rivals-only sample (3)
<i>BTM firm</i>	0.026*** (4.140)	0.026*** (4.150)	
<i>BTM rival</i>	-0.001** (-2.186)		
<i>SMST rival</i>		0.001 (0.360)	0.011** (2.115)
<i>SM rival</i>		-0.002* (-1.912)	
Observations	172,589	172,589	12,385
<i>R-squared</i>	.080	.080	.106
Firm FE	Yes	Yes	No
Event-day FE	Yes	Yes	Yes

This table presents OLS regressions of cumulative abnormal returns (CARs) around BTM announcements. The sample includes BTM firms, rival firms and control firms. CARs are based on a market model with parameters estimated during trading days $[-271, -21]$, relative to the BTM grant announcement date. The dependent variable (CAR) is actual return minus market-model-predicted return, cumulated over the 3 trading days $[-1, +1]$ surrounding the BTM award announcement. CARs are winsorized at the 1% and 99% levels. BTM firms are those that received the BTM award on the event date. Control firms are firms with products that do not fall in the ICD-10 market that experienced BTM entry on the event date.

BTM rival is an indicator equal to one for firms that have any drug project that targets an ICD-10 market and that experienced BTM entry on the event date. *SMST*, an abbreviation for same market same technology, is an indicator equal to one for rival firms that were shocked by BTM entry to a market where at least one of the rival's drug projects uses a technology (i.e., target action) that is the same technology used by the BTM-awarded drug. *SM* is an indicator equal to one for rival firms shocked by BTM entry in a product market where that rival's projects do not share any technological similarities with the BTM-awarded project. Column 1 and 2 present regression results for the full sample. Column 3 presents regression results for the sample of rival firms only. In column 3, the sample drops BTM firms completely from the quarter in which they received the BTM designation and until the end of the sample (this step helps avoid any confounding effects that may arise when subsequent BTM awards use similar technology as that used by the former BTM). In total, the tests below are based on the announcement returns of 957 unique firms around 343 BTM announcements. Standard errors are clustered by firm. *t*-statistics are reported in parentheses. *** $p < .01$; ** $p < .05$; * $p < .1$.

5.4 Table 3: Announcement returns of BTM firms, rivals, and controls

Read:

- BTM recipients gain about 2.6% around the announcement.
- Rivals lose value on impact, and the negative reaction is concentrated among SM rivals rather than SMST rivals.
- Within the rival sample, SMST firms outperform SM firms by about 1.1%, which already hints that technological closeness matters.

5.5 Table 4: Firm-level innovation after BTD shocks

Read:

- Despite the strong market reaction, rival firms do not show a robust firm-wide increase in patenting or R&D.
- The only clearer pattern is that SMST rivals reduce patenting, which the paper interprets as one possible reallocation effect after receiving a credible external signal about their technology.
- This table is the smoking gun for the aggregation problem inside pharma itself.

Table 4
BTB shocks and innovation at pharma firms: Firm-level aggregation concerns within pharma

	Patent dummy		Citation-weighted patents		R&D/TA	
	Full sample	Rivals only	Full sample	Rivals only	Full sample	Rivals only
	(1)	(2)	(3)	(4)	(5)	(6)
BTB firm	-0.036*	-0.031	-10.615	-6.865	-0.012	-0.013
	(-1.816)	(-1.569)	(-0.287)	(-0.176)	(-0.693)	(-0.746)
BTB rival	0.002		4.192		-0.012	
	(0.407)		(1.287)		(-1.458)	
SMST rival		-0.045**	-0.023	-35.906**	0.004	0.001
		(-3.197)	(-1.465)	(-1.987)	(0.327)	(0.049)
SM rival		0.004		5.667	-0.010	
		(0.675)		(1.621)	(-1.117)	
Observations	35,951	35,951	10,431	35,951	10,431	25,901
R-squared	.493	.552	.644	.643	.194	.194
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Calendar Qr FE	Yes	Yes	Yes	Yes	Yes	Yes

The tests in this table use OLS regressions to examine firm-level innovation following BTB events. BTB firm is an indicator equal to one for firms that received a BTB award in one of the previous 4 quarters. BTB rival is an indicator equal to one for firms developing operating in markets that experienced BTB entry in one of the last 4 quarters. SMST is an indicator equal to one for rival firms that were shocked by BTB entry in the previous four quarters in a market where at least one of the rival's drug projects uses a technology (i.e., target action) that is the same technology used by the BTB-awarded drug. SM is an indicator equal to one for rival firms shocked by BTB entry in the previous 4 quarters in a product market where that rival's projects do not share any technological similarities with the BTB-awarded projects. In columns 1-3, the dependent variable is the citation-weighted number of patents; in columns 4-6, the dependent variable is a dummy equal to 1 if the firm had issued a patent in the corresponding quarter; and in columns 7 and 8, the dependent variable is equal to the quarterly R&D expense divided by total assets. The full sample is included where the column heading indicates "Full sample," whereas only the sample of rivals is included where the column heading indicates "Rivals only." Standard errors are clustered by firm. t-statistics are reported in parentheses. ***p < .01; **p < .05; *p < .1.

Table 5
Summary statistics on the phase-II development sample

A. Summary statistics on rivals', and control firms': phase-II projects

Subsample	All projects			Projects ever shocked by BTB			Projects never shocked by BTB		
	All (1)	Low comp (2)	Hi comp (3)	All (4)	Low comp (5)	Hi comp (6)	All (7)	Low comp (8)	Hi comp (9)
N	195,736	85,846	109,890	97,900	20,974	76,926	97,836	32,964	64,872
# of projects	12,038	5,838	6,200	5,167	1,197	3,970	6,871	4,641	2,230
# of markets	902	659	104	185	168	55	807	791	49
# of firms	2,319	1,698	1,540	1,319	906	1,116	1,000	792	424
% graduated	12.97	13.98	10.01	7.57	14.61	5.34	16.59	16.33	18.38
Competition	4.34	3.11	5.53	4.97	3.51	5.566	3.61	2.92	5.44
Market exp %	12.19	12.08	12.51	11.17	10.36	11.51	13.38	12.88	14.73

B. Summary statistics for rivals' phase-II projects conditional on the type of rival

Subsample	Rival SM projects			Rival SMST projects		
	All (1)	Low comp (2)	Hi comp (3)	All (4)	Low comp (5)	Hi comp (6)
N	55,247	11,531	43,716	5,816	1,195	4,621
# of projects	4,730	1,092	3,638	526	114	412
# of markets	183	153	55	86	51	41
# of firms	1,254	582	991	270	81	221
% graduated	6.79	14.11	4.58	15.26	25.83	12.36
Competition	5.21	3.63	5.68	5.22	3.52	5.68
Market exp %	10.78	11.19	10.65	13.11	8.28	14.45

This table presents summary statistics for phase-II projects, therapeutic markets, and developer firms appearing in the phase-II development sample. Phase-II projects owned by BTB awarded firms are dropped from the sample if they target the same market in which the BTB firm was awarded. The observation level in the sample is project-quarter and is displayed in the first row of each panel for each corresponding sample. In both panels below, the variables # of projects, # of markets, # of firms display the unique number of phase-II projects, number of therapeutic markets and number of firms developing phase-II projects in each corresponding sample, respectively. % graduated reports the percentage of all phase-II projects that advance to phase-III in the corresponding sample. Competition displays the average number of competing rival projects in the same market. Market exp % is the average exposure of a firm in a market and is calculated as the number of projects owned by a firm in that market divided by the total number of projects owned by a firm. In both panels below, columns titled "All" include all phase two projects in the corresponding sample. Whereas columns titled "Low comp" ("Hi comp") use the subsample of phase two projects in markets with low (High) competition, that is, markets with competition levels below (above) the median level of competition in the full sample.

In panel A, the first three columns are based on the full sample; columns 4-6 are based on the subsample of rival phase-II projects that have ever experienced a BTB shock and columns 7 and 8 are based on the subsample of control phase-II projects that have never experienced BTB entry. In panel B, the sample focuses only on rival phase-II projects. The first 3 columns display the corresponding statistics for rival SM phase-II projects that experience BTB entry but use technology that is different from that used by the BTB-awarded drug. In columns 4-6 of panel B, statistics are displayed for rival SMST phase-II projects that experience BTB entry and use the same technology that was used by the drug that was awarded a BTB in the same market.

5.6 Table 5: Summary statistics for the phase-II sample

Read:

- BTB-shocked phase-II projects are much more concentrated in high-competition markets than control projects.
- SMST rivals have noticeably higher graduation rates than SM rivals, especially in low-competition markets.
- These descriptives already foreshadow the two core forces in the paper: competition and distance to frontier.

5.7 Figure 2: Event-time hazard coefficients for all rival projects

Read:

- Pretrends are flat, which is crucial for the design.
- Post-shock divergence is strongest in high-competition markets, where rival continuation falls, and in low-competition markets, where it rises.
- This is the dynamic visual counterpart to the inverted-U logic later quantified in Table 6.

5.8 Figures 3 and 4: Event-time hazards for SMST and SM projects

Read:

- Figure 3 shows that SMST projects are generally more likely to continue after a BTB shock.

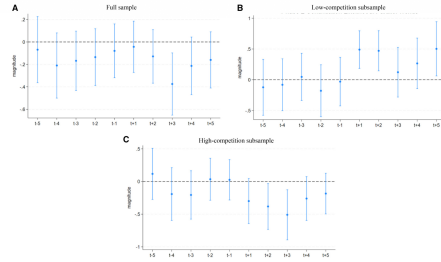


Figure 2
Coefficients from a Cox proportional hazards model that estimates the graduation (to phase-III) rates of rival phase-II projects relative to control projects, in the 5 years before and 5 after, BTD entry into a therapeutic market
The success event, *Graduation*, is an indicator equal to one in the quarter that a project graduates to phase-III, and equal to zero in all quarters where it remains in phase-II. The analysis time is the number of quarters since the start of phase-II. The analysis sample is summarized in Table 5. The plotted coefficients are interaction variables between an indicator, ever-BTD shocked, which equals one if a phase-II project ever experiences BTB entry, with annual event time indicators for each of the 5 years before and after BTB entry. The bar caps cover the 95% confidence intervals for each coefficient. The Cox proportional hazards model is stratified by therapeutic market, includes calendar quarter fixed effects, and clusters standard errors by drug project. The analysis sample in Figure 2A is the full sample of phase-II projects. The sample in panel B (panel C) includes phase-II projects in markets with low (high) competition, that is, markets with competition levels below (above) the median competition level in the full phase-II sample.

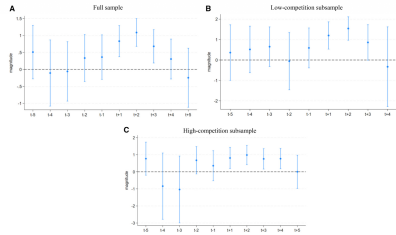


Figure 3
Coefficients from a Cox proportional hazards model that estimates the graduation (to phase-III) rates of rival SMST phase-II projects relative to control projects, in the 5 years before and 5 after, BTB entry into a therapeutic market
The success event, *Graduation*, is an indicator equal to one in the quarter that a project graduates to phase-III, and equal to zero in all quarters where it remains in phase-II. The analysis time is the number of quarters since the start of phase-II. The analysis sample is summarized in Table 5. The plotted coefficients are interaction variables between an indicator, ever-SMST with annual event time indicators for each of the 5 years before and after BTB entry. Ever-SMST is an indicator that equals one if a rival's phase-II project ever experiences the entry of a BTB drug in its market that uses the same technology (i.e., the rival's phase-II project shares at least one target action with the BTB drug). The bar caps cover the 95% confidence intervals for each coefficient. The Cox proportional hazards model is stratified by therapeutic market, includes calendar quarter fixed effects, and clusters standard errors by drug project. The analysis sample in Figure 3A is the full sample of phase-II projects. The sample in panel B (panel C) includes phase-II projects in markets with low (high) competition, that is, markets with competition levels below (above) the median competition level in the full phase-II sample.

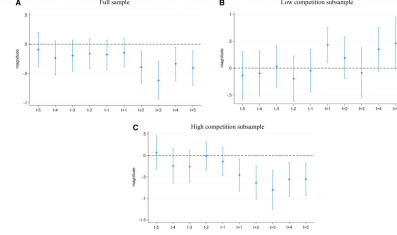


Figure 4
Coefficients from a Cox proportional hazards model that estimates the graduation (to phase-III) rates of rival SM (only) phase-II projects relative to control projects, in the 5 years before and 5 after, BTB entry into a therapeutic market
The success event, *Graduation*, is an indicator equal to one in the quarter that a project graduates to phase-III, and equal to zero in all quarters where it remains in phase-II. The analysis time is the number of quarters since the start of phase-II. The analysis sample is summarized in Table 5. The plotted coefficients are interaction variables between an indicator, ever-SM with annual event time indicators for each of the 5 years before and after BTB entry. Ever-SM is an indicator that equals one if a rival's phase-II project ever experiences the entry of a BTB drug in its market that uses a different technology (i.e., the rival's phase-II project uses different target actions than those used by the BTB drug). The bar caps cover the 95% confidence intervals for each coefficient. The Cox proportional hazards model is stratified by therapeutic market, includes calendar quarter fixed effects, and clusters standard errors by drug project. The analysis sample in Figure 4A is the full sample of phase-II projects. The sample in panel B (panel C) includes phase-II projects in markets with low (high) competition, that is, markets with competition levels below (above) the median competition level in the full phase-II sample.

- Figure 4 shows that SM projects obey the competition story more sharply: low-competition markets encourage catch-up, high-competition markets discourage it.
- Together, the two figures visually separate the Hashmi distance-to-frontier channel from the Aghion competition channel.

5.9 Table 6: Main hazard results

Table 6
Phase-II development and BTB shocks

	A. Product-level tests of BTB shocks and phase-II development											
	Full sample (any exposure)				Low market exposure				High market exposure			
	Any cmp (1)	Lo cmp (2)	Hi cmp (3)	Hi cmp (4)	Any cmp (5)	Lo cmp (6)	Hi cmp (7)	Hi cmp (8)	Any cmp (9)	Lo cmp (10)	Hi cmp (11)	Hi cmp (12)
BTB shock	0.013 (0.181)	1.173*** (5.498)	0.426*** (4.101)	-0.265*** (-2.718)	0.099 (1.026)	0.803*** (2.543)	0.315*** (2.254)	-0.042 (-0.314)	-0.087 (-0.837)	1.408*** (4.691)	0.558*** (3.484)	-0.517*** (-3.602)
BTB shock*Comp		-0.245*** (-5.545)				-0.149*** (-2.244)				-0.316*** (-5.159)		
Competition		0.005 (0.218)				0.004 (0.119)				-0.002 (-0.060)		
Observations	195,736	195,736	97,900	97,836	97,868	97,868	51,763	46,105	97,868	97,868	46,137	51,731

(continued)

Table 6
Continued

	B. Product-level tests of BTB shocks and phase-II development conditional on the type of rival											
	Full sample (any exposure)				Low market exposure				High market exposure			
	Any cmp (1)	Lo cmp (2)	Hi cmp (3)	Hi cmp (4)	Any cmp (5)	Lo cmp (6)	Hi cmp (7)	Hi cmp (8)	Any cmp (9)	Lo cmp (10)	Hi cmp (11)	Hi cmp (12)
SMST	0.621*** (5.132)	1.238*** (2.720)	0.860*** (4.460)	0.470*** (2.952)	0.517*** (2.857)	0.382 (0.454)	0.500** (1.677)	0.491** (2.076)	0.709*** (4.311)	1.869*** (4.125)	1.248*** (4.900)	0.459*** (2.070)
SM	-0.101 (-1.348)	1.243*** (5.590)	0.363*** (3.332)	-0.431*** (-4.112)	0.024 (0.238)	0.922*** (2.838)	0.288** (1.977)	-0.153 (-1.096)	-0.245*** (-2.185)	1.432*** (4.455)	0.462*** (2.755)	-0.751*** (-4.768)
SMST*Competition		-0.127 (-1.512)				0.027 (0.165)				-0.253** (-2.558)		
Competition		-0.266*** (-6.088)				-0.191*** (-2.783)				-0.358*** (-5.396)		
Observations	195,736	195,736	97,900	97,836	97,868	97,868	51,763	46,105	97,868	97,868	46,137	51,731

The tests in this table examine the likelihood of phase-II development following BTB shocks. The results displayed below are from the main Cox Proportional Hazards with estimates are stratified by market and the analysis time is equal to the number of quarters since the start of phase-II trials. The success event, *Graduation*, is an indicator equal to one in the quarter that a phase-II project graduates to phase-III, and equal to zero in all other quarters. The analysis sample is summarized in Table 5. Competition is calculated each quarter as the natural logarithm of the total number of drug projects in a therapeutic market. In both panels below, the full sample of phase-II projects is used in columns 1 and 2, and in columns 3 and 4, the full sample is partitioned by the level of competition: projects in markets below (above) the median level of competition are used in columns 5 (columns 6). Whereas the samples used in columns 5-12 are partitioned by market exposure. Market exposure is calculated by dividing the number of a firm's projects that target the focal drug project's market by the total number of projects owned by the same firm, in each quarter. The results displayed in columns 5-8 (columns 9-12) are based on the subsample of projects that fall below (above) the sample's median level of market exposure. In addition, columns 7 and 11 (columns 8 and 12) restrict the corresponding sample to projects in markets with competition levels below (above) the median level of competition in the full sample.

In panel A, the variable *BTB shock* is an indicator equal to one for all phase-II projects that reside in a market that had experienced BTB entry, and equal to zero otherwise. In panel B, the variable *SM* is an indicator equal to one if the focal project targets a market that had experienced BTB entry and but uses a different technology, than that used by any of the drugs that were awarded a BTB in that market. SMST is an indicator equal to one if the focal project had experienced BTB entry into its market and uses at least one target action (i.e., technology) that is also used by any of the drugs that were awarded a BTB in the same market. Standard errors are clustered by drug project z -statistics are reported in parentheses. *** $p < .01$; ** $p < .05$; * $p < .1$.

Read:

- In the full sample, the unconditional BTB-shock coefficient is near zero, but the interaction with competition is negative and highly significant (-0.245).
- Splitting the sample shows the inverted-U clearly: in low-competition markets the BTB shock raises continuation hazards (coefficient about 0.426, roughly a 50% increase), while in high-competition markets it lowers them (coefficient about -0.265 , roughly a 23% decline).

Table 7
Phase-II development around BTd shocks in ex ante leveled markets

	Full sample		Market competition	
	(1)	(2)	Low (3)	High (4)
<i>A. Ex ante leveled: Markets with no approved-for-sale products</i>				
	(1)	(2)	(3)	(4)
<i>BTd shock</i>	0.446** (2.197)	1.405*** (3.542)	1.091*** (5.279)	-2.210** (-2.184)
<i>BTd shock*Competition</i>		-0.225** (-2.180)		
Observations	24,884	24,884	20,693	4,191
<i>B. Ex ante leveled: Markets with no approved-for-sale products conditional on rival type</i>				
	(1)	(2)	(3)	(4)
<i>SM</i>	0.310 (1.352)	1.226*** (2.717)	0.931*** (3.977)	-2.023** (-2.005)
<i>SMST</i>	1.019*** (2.653)	1.831*** (2.586)	1.783*** (4.614)	-33.435 (0.000)
<i>SM*Competition</i>		-0.209* (-1.856)		
<i>SMST*Competition</i>		-0.206 (-1.111)		
Observations	24,884	24,884	20,693	4,191

The tests in this table examine the effect of BTd shocks on the development likelihood of phase-II projects conditional on the ex ante levelness of the market. The table presents coefficients from the main Cox Proportional Hazards specification, which stratifies the sample by market and clusters standard errors by drug project. The model's success event, Graduation, is an indicator equal to one in the quarter that a project graduates to phase-III, and equal to zero in all quarters where it remains in phase-II. The analysis time is the number of quarters since the start of phase-II development. The analysis sample includes phase-II projects in ex ante level markets where none of the competing firms own an approved-for-sale product (i.e., no approved products are on the market). Competition is calculated each quarter as the natural logarithm of the total number of drug projects in a therapeutic market. In both panels below, the full sample of phase-II projects is used in columns 1 and 2, and in columns 3 and 4, the full sample is partitioned by the level of competition; projects in markets below (above) the median level of competition are used in column 3 (column 4).

- Exposure matters: the pattern is much stronger for rivals whose project portfolio is concentrated in the shocked market.
- In panel B, SMST projects continue more aggressively, whereas SM projects retain the inverted-U pattern.

5.10 Table 7: Ex ante leveled markets (no approved-for-sale products)

Read:

- This is the paper's closer approximation to the neck-and-neck state in Aghion et al. (2005).
- The low-competition positive response is stronger here than in the full sample, while the high-competition response is sharply negative, albeit noisier because the subsample is small.
- The message is that the inverted-U is not an artifact of mature markets with entrenched incumbents; it also appears where the market starts closer to symmetric rivalry.

5.11 Table 8: Resource reallocation

Table 8
Potential reallocation of resources: Unshocked phase-II projects and new drug initiations

	<i>A. Rival development of unshocked Phase-II projects</i>			
	Low focal mkt competition (below sample median)		High focal mkt competition (above sample median)	
	(1)	(2)	(3)	(4)
<i>HCComp_HIEap shock</i>	0.254* (1.679)	0.479*** (2.762)	-0.335** (-2.296)	-0.278* (-1.744)
<i>HIEap shock</i>		0.167 (0.852)	-0.551** (-2.449)	-0.551** (-2.449)
<i>HCComp shock</i>		-0.833*** (-3.528)	-0.088 (-0.382)	-0.088 (-0.382)
Observations	22,783	22,783	22,770	22,770

The tests in panel A examine the development likelihood of never-shocked phase-II projects owned by active rivals that had recently experienced BTd entry in a market where they were developing a phase-II project. The results are based on the main Cox hazards model specification, which stratifies estimates by market and uses the analysis time in number of quarters since the beginning of phase-II. The sample includes never-shocked phase-II projects for active rivals. Active rivals are firms that have recently experienced BTd entry in a market in which they operate were developing a phase-II project. These variables are used in the tests below to identify the type of rival. *HCComp shock* is a dummy variable equal to one if the rival had experienced BTd entry in a highly competitive market (i.e., with a competition level that is above the median competition level in the full phase-II sample), and equal to zero if they experienced a BTd shock in a low competition market. *HIEap shock* is a dummy variable equal to one if the rival had experienced BTd entry in a market that the rival was highly exposed to (i.e., the rival's exposure was above the median level of firms' market exposure in the full phase-II sample), and is equal to zero if the rival had low exposure to the shocked market. *HCComp_HIEap shock* is a dummy variable equal to one for rivals that had experienced BTd entry in a competitive market and where they were highly exposed. The tests reported in columns 1 and 2 (columns 3 and 4) are based on the subsample of rivals' never-shocked projects in low (high) competition markets with a competition level below (above) the median level of competition in the full phase-II sample. *t*-statistics are reported in parentheses. *** $p < .01$, ** $p < .05$, * $p < .1$.

Table 8
Continued

Initiation type	Full sample			High exposure rival subsample		
	Initiation (1)	LowComp init. (2)	HIComp init. (3)	Initiation (4)	LowComp init. (5)	HIComp init. (6)
<i>HCComp_HIEap shock</i>	0.644*** (3.125)	0.012* (1.714)	0.005 (0.811)			
<i>HIEap shock</i>	-0.041*** (-3.491)	-0.021** (-2.286)	-0.028*** (-3.525)			
<i>HCComp shock</i>	0.018 (1.391)	0.007 (0.696)	0.012 (1.338)	0.038** (2.337)	0.032*** (2.968)	0.021* (1.976)
Observations	18,984	18,984	18,984	9,768	9,768	9,768
R-squared	.247	.203	.220	.216	.145	.163

The tests in panel B examine the initiations of rivals that had recently experienced BTd entry to a market where they were developing a Phase-II project. The table reports OLS estimates of drug initiations conditional on the type of rival. The sample used in the tests below has an observational level of firm-quarter and includes rivals that had recently experienced BTd entry in a market where they were developing a phase-II project. The analysis sample is firm-quarter. The dependent variables are *Initiation*, *LowComp init.*, and *HIComp init.*, and are defined as indicator variables equal to one in the quarter that a rival initiates a drug project, initiates a drug project in a low competition market (below the full drug sample's median of competition level) or initiates a drug project in a high competition market (above the full drug sample's median level of competition), respectively, and is equal to zero otherwise. *HCComp shock* is a dummy variable equal to one if the rival had experienced BTd entry, in one of the last four quarters, in a highly competitive market (i.e., with a competition level that is above the median competition level in the full phase-II sample), and equal to zero otherwise. *HIEap shock* is a dummy variable equal to one if the rival had experienced BTd entry, in the previous 4 quarters, in a market that the rival was highly exposed to (i.e., the rival's exposure was above the median level of firms' market exposure in the full phase-II sample) and is equal to zero otherwise. *HCComp_HIEap shock* is a dummy variable equal to one if for rivals that had experienced in the previous 4 quarters BTd entry in a competitive market and where they were highly exposed. In the first four columns below, the full sample of active rivals' are included. In the last four columns below, the sample is restricted only to rivals that were highly exposed to the market in which they had experienced BTd entry. All regressions include firm and quarter FE and SE clustered by firm. *t*-statistics are reported in parentheses. *** $p < .01$, ** $p < .05$, * $p < .1$.

Read:

	Acquisition (1)	Cash/AT (2)	SG&A/Rev (3)	Leverage (4)	Acquisition (5)	Cash/AT (6)	SG&A/Rev (7)	Leverage (8)
<i>A. Full sample of rivals and controls</i>								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>BTD rival</i>	-0.026*** (-3.521)	0.014 (1.405)	-1.639 (-0.142)	-0.046 (-0.391)	-0.038*** (-5.203)	0.024** (2.229)	-7.088 (-0.465)	-0.088 (-0.703)
<i>App BTD rival</i>					0.079*** (4.975)	-0.062*** (-4.484)	10.882 (1.295)	0.289*** (4.002)
Observations	26,316	26,143	7,319	26,018	26,316	26,143	7,319	26,018
R-squared	.721	.585	.662	.502	.721	.585	.662	.502
<i>B. Rivals-only sample</i>								
	Four quarters after rival is shocked				Eight quarters after rival is shocked			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>App BTD rival</i>	0.082*** (5.667)	-0.032*** (-2.673)	-15.093 (-1.392)	0.214*** (3.942)	0.103*** (7.436)	-0.037*** (-3.366)	-9.493 (-1.308)	0.202*** (4.233)
Observations	8,533	8,509	2,023	8,476	10,861	10,834	2,568	10,790
R-squared	.869	.693	.794	.439	.860	.678	.703	.349

This table displays results from OLS regressions of firm-level financial outcomes following BTD shocks. The sample used in this table includes firms in the full drug sample that matched with the Compustat quarterly fundamentals files. The sample has an observation level of firm-quarter and includes financial information on 1,109 firms in 50 quarters between 2010q1 and 2021q4. The dependent variables are indicated in each column heading. Acquisition is a dummy equal to one if the firm was involved in the acquisition of another firm or drug, in the corresponding quarter. Cash/AT and SG&A/Rev are a firm's quarterly cash holdings divided by total assets, and quarterly sales, general and administrative expenses divided by quarterly sales, respectively. Leverage is equal to total debt divided by total assets. All financial data have been winsorized at the 1% level.
In panel A, the full sample is used, and *BTD shock* is an indicator equal to one if the firm experienced BTD entry in the previous 4 quarters, and zero otherwise. *App BTD shock* is a dummy equal to one if the firm experienced BTD entry in the previous 4 quarters in a market where it has an approved for sale product.
In panel B, the sample is restricted to active rivals that have experienced BTD entry in a market where they were developing a drug project in the previous 4 quarters (columns 1-4) or 8 quarters (columns 5-8). All tests below include firm and year FE and cluster SE by firm. *t*-statistics are reported in parentheses. *** $p < .01$; ** $p < .05$; * $p < .1$.

- Panel A asks whether rivals accelerate unshocked phase-II projects elsewhere in the portfolio. The answer is yes, especially when the rival was highly exposed to a BTD shock in a high-competition market.
- Panel B studies new project initiations. Rivals start more projects after BTD shocks, and among highly exposed rivals this push is strongest toward low-competition target markets.
- The table therefore turns the aggregation argument into a concrete mechanism: firms shift effort across projects and toward softer markets.

5.12 Table 9: Financial outcomes after BTD shocks

Read:

- Not all rivals react the same way. Those without an approved drug in the shocked market retrench: fewer acquisitions, higher cash, weaker offensive response.
- Rivals with an approved-for-sale drug in the shocked market fight back: they acquire more, draw down cash, and lever up.
- This is a useful complement to the innovation evidence because it shows how firms finance and organize their response to competitive shocks.

6 Short summary

- **Conclusion.** Once innovation is measured at the project level, competition and innovation follow an inverted-U in the pharma data: rivals accelerate in less competitive markets and decelerate in more competitive markets after a BTD shock.
- **Mechanism.** The response depends not only on market competitiveness but also on technological distance to the new frontier. Projects that share technologies with the BTD drug are more likely to continue.
- **Empirical lesson.** Firm-level innovation measures can be badly misleading when firms reallocate effort across projects, so project-level data are essential for this question.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The paper establishes that the competition-innovation relation is not uniformly positive or negative; it depends on ex ante market competitiveness and on distance to the frontier.
- **Quantitative evidence.** A BTD shock raises rival phase-II continuation by about 50% in less competitive markets but lowers it by about 23% in more competitive ones.
- **Credibility.** The design is persuasive because BTD awards are salient, market-specific, and early-stage signals; stock returns move sharply on impact; and event-study pretrends are flat.
- **Mechanism evidence.** SMST projects, which are closer to the new frontier, are more likely to continue, consistent with the Hashmi extension of Aghion et al. (2005).
- **Aggregation lesson.** Both the manufacturing benchmark and the firm-level pharma results show that broad observation units conceal the underlying causal response.
- **Portfolio response.** Rivals do not merely stop; they reallocate. Unshocked phase-II projects and new low-competition initiations pick up, and firms with incumbent products in the shocked market actively defend their position.

7.2 Economic intuition

- A BTD shock reduces the relative standing of incumbent rivals in one therapeutic market. When that market is not very competitive, there is still room to earn rents by catching up, so rivals work harder on their projects.
- When the market is already highly competitive, the rents from catching up are smaller; the same unleveling shock therefore makes further innovation less attractive.
- Rivals that already use the new leading technology are not as far behind. For them, the shock is less discouraging and can even be motivating because the path to catch-up is shorter.

7.3 Takeaway

This is a strong paper because it does not just re-estimate the competition-innovation relationship in a new setting; it redesigns the measurement problem. By moving from firms to projects and by using BTD awards to create clear therapeutic-market shocks, it shows why the literature has been so mixed and what the underlying mechanism actually looks like. The bottom line is that competition matters, but only once we know *where* inside the firm the shock hits and *how far* the rival is from the new technological frontier.